

Apart from Bengaluru that features in the list of top locations for jobs in embedded field with 32%, figures indicate Delhi, Mumbai and Pune share 20%, 15% and 12% of the jobs, respectively. Remaining jobs are scattered in other locations.

Low pay initially

Talking about Tevatron, Anurag Gupta, director, Tevatron Technologies informs, "We recruit experienced and fresh engineers for embedded systems, firmware, VLSI design and chip design. The package varies from candidate to candidate, ranging from 300,000 to 2,000,000 rupees per annum, based on their experience level."

At Eleics Design, fresh graduates spend their initial months in training and learning, and hence they offer them a package of 100,000 rupees per annum. Solely based on their performance, pay is gradually incremented, informs Gohel. Talking about experienced professionals in embedded design, he says, "They will get a pay package of around 200,000 to 320,000 rupees per annum and, again based on performance, s/he will receive increments and other perks." He adds, "In some cases, we also pay on hourly basis to experts, and that amount varies as per his/her design specific expertise."

Vedi informs, "VVDN Technologies hires B. Tech/B.E. graduates from electronics, computer science, IT and electrical branches. For freshers, an approximate pay of 240,000 rupees is provided whereas for experienced professionals, it depends on their experience and knowledge."

At Caliber Embedded Technologies, Sakthivel says, "For fresh graduates we are offering between 125,000 and 175,000 rupees per annum and for experienced professionals we are offering 250,000 to 350,000 rupees per annum, based on the level of experience of the candidate."

On behalf of elfnfochips, Kapat too affirms that they hire fresh engineers from engineering institutes all across the country. She says, "Depending on their qualification and institute, sal-

Do some institutes have a greater edge?

► "We recruit from IITs/NITs as well as other institutions. In particular, for freshers, we are looking for strong concepts and fundamentals rather than the college they come from." —**Anurag Gupta, director, Tevatron Technologies**

► "We need skillful engineers, we do not see IIT/NIT tags and sometimes we even do not see their mark sheets." —**Amrit Gohel, founder, Eleics Design**

► "We are not restricted to IITs/NITs alone but do head hunting in other engineering institutions as well, as we believe that right talent is not restricted to premier institutions." —**Girija Vedi, senior marketing executive, VVDN Technologies**

► "We focus on good local engineering institutes and hire based on raw talent and train them for our needs. For small companies, it is difficult to get placement slots at IITs/NITs. There are several exceptional IITians I have engaged with in past and had great experience working with them. But, I have mostly come across research oriented and analytical IITians, who although intelligent and bright, were not able to tackle real life design problems. So, for normal engineering designs, our preference is for engineers from other institutes." —**Upendra Patel, CEO, Glide Technology Pvt Ltd**

► "We do not have any such reservations about the institutions, but it is true that being a small-scale company, we would do very stringent screening to ensure that people we recruit can ramp up quickly to our expectations and start delivering at the earliest. It is simply because we can do have much bandwidth to provide long trainings after the recruitment and bridge the gap." —**Vinod D. Halappa, senior manager - business development, Dexcel Electronics Designs Pvt Ltd**

ary can be in the range of 200,000 to 300,000 rupees per annum."

At Glide Technology, Patel informs, "We have a mix of senior persons with freshers. Freshers' salaries range from 120,000 to 180,000 rupees and largely depend on the institution they had studied at and their college project. For experienced professionals, it is based on past experience, and there are no fixed criteria. We even hire freelance professionals with specialised skills on hourly rates for specific jobs."

Skills expectation

The industry expects basic knowledge and skills from the freshers but specific skills from the experienced. For the skills expected from a fresher, see the box titled 'Expectations from freshers,' which is based on views of the industry experts.

The skills expected from experienced candidates are much more specific, apart from the basics and soft skills of course. Tiwari says, "They should be equally conversant with hardware and software, particularly analogue IC, components and the digital counterparts, interfacing of I/O peripherals to microcontroller and building the embedded core programming logic from scratch." He adds, "They should be aware of the latest developments and emerging technologies like 32-bit ARM processor, Raspberry Pi, Arduino, Embedded

Linux, RTOS and interfacing Android apps to embedded hardware."

Speaking more specifically, Gohel says, "S/he must have worked on at least three MCU platforms such as Microchip PIC16/PIC18/PIC32, TI MSP430, Renesas RX62N and Atmel ATmega32 to list a few, and must be good at electronics and software development. S/he should have stabilised at least two or more products in the past and should also have a flair for PCB designing."

But when it comes to the experienced graduates, they are expected to carry product engineering/development knowledge for some specific domains like the consumer, defence, industrial automation, semiconductor, avionics or bio-medical sector, informs Halappa. He says, "In general, they must be strong in design aspects and considerations to meet the requirements for a product in a given domain in terms of performance, cost, quality and standard compliance. Specially quality and process compliance would be of great importance when it comes to defence, avionics and bio-medical products."

Experts' advice

Embedded systems, being so omnipresent and so inevitably embedded into our modern lifestyle, could be the single largest enabler that would play a significant and ever-growing role in shaping our future tools, gadgets,